

Chairperson: Voisin, Kevin

Secretary: Voisin, Kevin

Present:

Borgeat, Rémy

Carrino, Francesco

Voisin, Kevin

Schedule

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Start of meeting

1 Direction, Strategic: Anomaly Detection vs. Forecasting

The main discussion focused on pivoting the core methodology of the thesis from anomaly detection to forecasting.

- **Physical Correlation:** It has been theoretically and practically established that tool wear is directly correlated with the machine's spindle torque. As the cutting edge degrades, the required torque increases.
- **Methodology Pivot:** Instead of attempting to detect "anomalous" behavior without a clear failure point, the strategy will now focus on predicting (forecasting) the torque. By accurately forecasting the torque trajectory using sensor data, tool wear can be indirectly monitored and optimized.
- **Action Item:** From this point forward, all literature review and model testing should strictly focus on Time Series Forecasting approaches applied to the torque data.

2 Data Management and Processing

2.1 Data Handling with Disco

- **Transfer Limitations:** There are potential limitations regarding transferring the raw datasets to the Disco computing environment.
- **Action Item:** If data cannot be directly transferred to Disco, the required features and data transformations (e.g., spectrograms, statistical features) must be pre-calculated locally before uploading the processed datasets to the cluster.

2.2 Sensor Units

- **Clarification Needed:** The exact unit of measurement for the accelerometer data is currently unconfirmed (possibly $\frac{m}{s^2}$). This needs to be verified for accurate physical interpretation.

2026-05-26

HES-SO Valais-Wallis - Bachelor Thesis: Weekly Update Meeting

15:00 PM Meeting Adjourned

Reformulated with Gemini

Warnings:

Page 1: "Direction, Strategic" spoke, but is unaccounted for